

Spencer Hilligoss

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Education

PhD in Statistics

University of California, Irvine

Sep 2023 – Present
(Expected December 2027)

- **Coursework:** Advanced Probability, Machine Learning, Statistical Consulting
- **Research Topics:** Time-series Forecasting, Deep Learning, Generative Modeling, Causal Inference
- **Supervised by:** Dr. Annie Qu and Dr. Tianchen Qian
- **GPA:** 3.929/4.0

B.S. in Statistics and Operations Research, B.A. in Economics

University of North Carolina at Chapel Hill

Aug 2019 – May 2023

- **Coursework:** Probability, Statistical Machine Learning, Real Analysis, Stochastic Modeling, Dynamic Decision Making, Financial Markets, Micro and Macro Economic Theory
- **GPA:** 3.7/4.0

Fellowships and Awards

University of California, Irvine

- T32 STEER in Biomedical Sciences Fellow
- Diversity Recruitment Fellowship

Aug 2025 – Present

Aug 2023

Publications and Current Work

Hilligoss, S., Qu, A. (2026). “Mixed-Effects Denoising Diffusion for Heterogeneous Time Series.” *Under review*.

Hilligoss, S., Qu, A. (2026). “Causal Mediation Pathways in Continuous Postprandial Glucose Monitoring for Type 1 Diabetes Patients.” *npj Metabolic Health and Disease*, 4, 35. [doi:10.1038/s44324-026-00127-z](https://doi.org/10.1038/s44324-026-00127-z) 

Hilligoss, S., Saha, A., Qian, T. “Selective Inference for Time-Varying Causal Effect Moderation via the Ordered Lasso.” *Manuscript in preparation*.

Research and Work Experience

Mixed-Effects Denoising Diffusion for Heterogeneous Time Series

University of California, Irvine — under review

Mar 2026 – Aug 2026

- Led development of a dual-pathway diffusion model for heterogeneous time-series generation, mirroring the mixed-effects decomposition $y = X\beta + Zb$; generated samples matched subject-level distributions more closely than the PaD-TS baseline on every benchmark (clinical glucose, mobile sensing, equities).
- Validated results in peer review with multi-seed confidence intervals, ablations, and out-of-sample checks.

Sentiment-Enhanced LSTM Forecasting (DJIA)

University of California, Irvine

Sep 2024 – Dec 2024

- Engineered a daily feature set combining market data with text-based sentiment signals (news/social text) and aligned features to next-day index returns.
- Implemented and trained LSTM forecasting models; compared against baseline time-series models using rolling/expanding-window backtests and out-of-sample error metrics.

The Lasting Impact of HOLC Redlining on Minorities in New York

Harvard University, Research and Policy Institute

June 2022 – Aug 2022

- Built a reproducible R workflow linking HOLC redlining maps to demographic data; estimated regression-based effects of historical redlining on present-day outcomes with robustness and sensitivity checks.

Mentorship

Undergraduate Research Mentor, University of California, Santa Barbara

- Weekly research meetings with two undergraduates: financial time-series modeling and generative diffusion modeling applied to soccer tactics.

Computing Tools

Python (PyTorch, NumPy, Pandas, scikit-learn); R; Git; Linux; HPC/Slurm; Stata; Excel